National Technical University of Ukraine
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Faculty of Informatics and Computer Science
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LINUX

INSTRUCTIONS FOR PERFORMING LABORATORY WORKS FOR STUDENTS

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LABORATORY WORK #6

Topic

User identification. Umask in Linux. Hard and symbolic links.

Tasks

1) Familiarize yourself with the id, whoami, chown commands for a file/directory/link. Learn to use them.
2) Familiarize yourself with the Umask command. Create file and directory, view
umask, explain. Change umask for newly created files.
3) Familiarize yourself with hard and symbolic links. Create a hard and symbolic
link. Explain the difference.

Task 1.1: Familiarize yourself with id, whoami, chown commands.
To view user and group information:

```
mehmet@mehmet-VirtualBox:~$ id
uid=1000(mehmet) gid=1000(mehmet) gruplar=1000(mehmet),4(adm),24(cdrom),27(sudo)
,30(dip),46(plugdev),122(lpadmin),135(lxd),136(sambashare),1002(memo_group)
```

This will display the current user's UID, GID, and groups.
To view the current username:

```
mehmet@mehmet-VirtualBox:~$ whoami
mehmet
```

Task 1.2: Familiarize yourself with the Umask command.
View the current umask:
To change the owner of a file/directory:

```
mehmet@mehmet-VirtualBox:~$ sudo chown mehmet_user:memo_group file
[sudo] mehmet için parola:
mehmet@mehmet-VirtualBox:~$ umask
0002
```

The output will be in octal format. Create a file and directory:

```
mehmet@mehmet-VirtualBox:~$ touch newfile
mehmet@mehmet-VirtualBox:~$ mkdir newdirectory
```

Change umask for newly created files:

```
mehmet@mehmet-VirtualBox:~$ umask 022
```

Task 2

Task 2.1: Familiarize yourself with hard and symbolic links.

Create a hard link:

```
mehmet@mehmet-VirtualBox:~$ ln -s file dosya
mehmet@mehmet-VirtualBox:~$ ls
%1                      dosya1_sorted.txt   Masaüstü           output.txt
1.txt                   example.txt         mehmet1.txt        Resimler
2.tct                   file                mehmet2.txt        snap
2.txt                   file1_sorted.txt    mehmet.txt         subdir
Belgeler                file1.txt           Müzik              Şablonlar
city_1                  file2_sorted.txt    my_archive.tar.gz  try.txt
city_country.txt        file2.txt           newdirectory       Videolar
destination_file.txt    Genel               newfile
dir1                    hellofile.txt       new_pass.txt
dosya                   İndirilenler        new.txt
mehmet@mehmet-VirtualBox:~$
```

```
mehmet@mehmet-VirtualBox:~$ touch directory
mehmet@mehmet-VirtualBox:~$ ln directory dosya
mehmet@mehmet-VirtualBox:~$ touch input.txt
mehmet@mehmet-VirtualBox:~$ ln input.txt dosya
mehmet@mehmet-VirtualBox:~$
```

This creates a hard link named hardlink pointing to existingfile.

Create a symbolic link:

```
mehmet@mehmet-VirtualBox:~$ ln -s directory dosya1
mehmet@mehmet-VirtualBox:~$ ln -s file dosya1
```

Hard Links:

● Essentially additional directory entries for the same inode.
● Changes to one hard link affect all others since they point to the same data blocks.
● Cannot link directories.
● Cannot link across file systems.

Symbolic Links:

● Points to the pathname of the target file or directory.
● Independent of the target, changes to the target won't affect the link.
● Can link directories.
● Can link across file systems.